SIMATSENGINEERING
ASSESSMENTTOOL1

COURSENAME: CSA16COURSECODE:DataWarehousingandDataMining

COURSEOUTCOMECOVERED

CO4: Examinetheapplicabilityofclusteringalgorithmsinreal-worldscenarios,presenting findings through clear reports with effective visualizations.
AssessmentToolWeightage: 40%

QUESTIONS:05 **TotalMarks:50**

AnnexureA

DataInterpretation

Criteria	Understanding of Data ***(2.5)***	Analysis***(2.5)***	Application of Concepts ***(2)***	Conclusion ***(1.5)***	Clarity***(1)***	***Total(10)***
Weightage	*25%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

CodeofConduct:

M. ***Jayaprakash(192425268)(****Name/RegNo)certifythatthissubmissionismyoriginalwork and that I have adhered to the guidelines specified for this assessment.*
Iunderstandthatanyviolationofacademicintegrityruleswillresultindisciplinaryaction.
Signatureofthestudent:
jayaprakash

RUBRICSFORCALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all datapatterns, trends, and relationships	Interpretsdata correctly with minor gaps	Partial understanding ; misses key trends	Misinterprets or fails to understanddata
Analysis	25%	Providesdeep analysis with strong logical reasoningand justification	Provides reasonable analysiswith some justification	Basicanalysis with weak reasoning	Nomeaningful analysis provided
Applicationof Concepts	20%	Effectively appliesrelevant concepts/tools tointerpretdata accurately	Applies conceptswith minor errors	Limited application ; requires guidance	Unabletoapply conceptstodata
Conclusion	20%	Drawsclear, meaningful conclusions with strong insightsand implications	Providesvalid conclusions withmoderate insights	Conclusions arevagueor weak	Noclearor valid conclusion s
Clarity	10%	Presents interpretatio n clearlyusing proper structure, visuals,and terminology	Generallyclear with minor issues	Lacksclarityor proper structure	Poor presentation anddifficultto understand

AnnexureA

Q1.Dataset:HospitalPatientSegmentation

PatientID	Age	MonthlyMedicalExpense(₹)	HealthRisk Score
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSEValues

K	SSE
1600	
2350	
3220	
4200	
5190	

Question(BL4–8Marks):

AhealthcareorganizationusesK-Means clusteringtogrouppatientsbased ontheirmedical expensesandhealthriskscores.InterprettheelbowpointfromtheSSEvaluesprovidedand determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

Q2.Dataset:E-CommerceOrderAnalysis

OrderID	ProductCategory	OrderValue(₹)
O1	Electronics	55000
O2	Clothing	25000
O3	Electronics	57000
O4	Furniture	30000
O5	Clothing	27000

Question(BL4–8Marks):

Anonlineretailerapplies hierarchicalclusteringto groupcustomerordersbasedonproduct category and order value. A dendrogram is generated from the dataset. Interpret how the clustersareformedandidentifytheappropriatenumberofclusters.Explainthesignificance of the selected clusters in understanding customer purchasing behavior.

Q3.Dataset:MobileAppUserBehavior

User	DailyUsageTime(min)	LoginFrequency	In-AppPurchase(₹)
U1	120	18	2000
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	130	20	1800

Question(BL5–8Marks):

AmobileapplicationcompanyusestheHACalgorithmtosegmentusersbasedontheir activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Q4.Dataset:AgriculturalCropMonitoring

Farm	Rainfall (mm)	Soil Fertility Score	Crop Yield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Question(BL4–8Marks):

Anagriculturalresearchcenterperformsclusteringonfarmdatabeforeandafterapplying normalization. Interpret the differences observed in the clustering results. Explain how normalizationinfluencesclusterformationandimprovesthereliabilityofagriculturaldata analysis.

Q5.Dataset:EmployeePerformanceEvaluation

Employee	ProductivityScore	Attendance(%)	SkillAssessmentScore
E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette Scores

Model	Score
K=2	0.48
K=3	0.68
K=4	0.52

Question(BL5–8Marks):

Acompanyclustersemployeesbasedonproductivity,attendance,andskillassessmentscores. Analyzethesilhouettescoresobtainedfordifferentclusteringmodelsanddeterminethemost suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

Answers

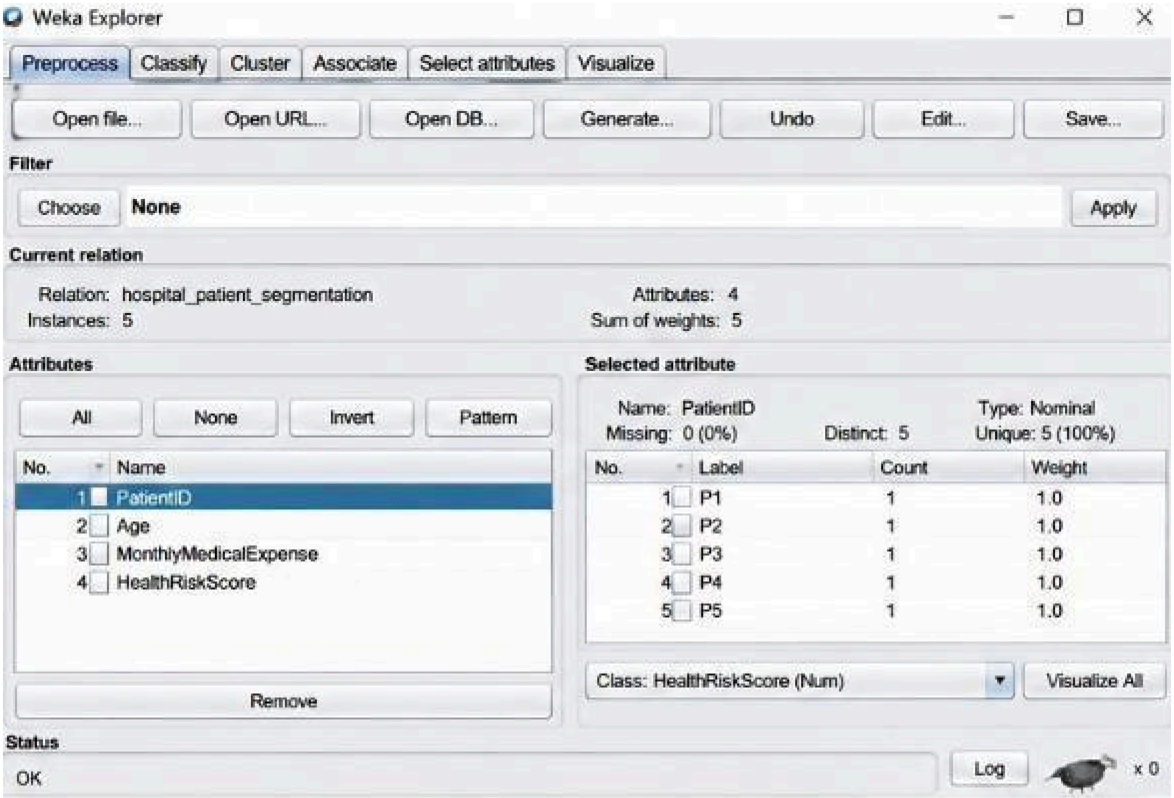
Q1.HospitalPatientSegmentation
Answer
ThegivenSSEvaluesare:

NumberofClusters(K)	SSE
1	600
2	350
3	220
4	200
5	190

IntheK-Meansalgorithm,SSE(SumofSquaredErrors)decreases as thenumberofclusters increases.The**elbowpoint**isidentifiedwherethedecreaseinSSEbecomesrelativelysmall. FromK=1toK=2,SSEdecreasesfrom**600to350**.FromK=2toK=3,itdecreasesfrom**350 to 220**. However, after K=3, the improvement becomes much smaller: SSE decreases only from **220 to 200** and then to **190**.
Therefore,theelbowoccursat**K=3**,sotheoptimalnumberofclustersis**3**. **Patient Segmentation**
Thepatientscanbedividedintothreegroupsbasedontheirmedicalexpensesandhealthrisk scores:

- **Cluster1–High-riskpatients:**Patientswithhigherhealthriskscoreswhomay require frequent monitoring and medical attention.
- **Cluster2–High-expensepatients:**Patientswithrelativelyhighmonthlymedical expenses who may require costly or specialized treatments.
- **Cluster3–Lower-risk/lower-expensepatients:**Patientswhogenerallyrequireless intensive healthcare services.

Importance
Selecting threeclustershelps thehealthcareorganization identify different patient groupsand providetargetedhealthcareservices.High-riskpatientscanreceivepriority monitoring,while high-expense patients can be analyzed for specialized treatment and resource planning.
Conclusion
Optimal number of clusters = 3. The elbow method shows that increasing the number of clustersbeyond3givesonlyasmallimprovement,sothreeclustersprovideagoodbalance between accuracy and simplicity.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Classify
Cluster
Associate
Select attributes
Visualize
Open file...
Open URL...
Open DB...
Generate...
Undo
Edit...
Save...
Filter
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Apply
Current relation
Relation: hospital_patient_segmentation
Instances: 5
Attributes: 4
Sum of weights: 5
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No. Name
1 PatientID
2 Age
3 MonthlyMedicalExpense
4 HealthRiskScore
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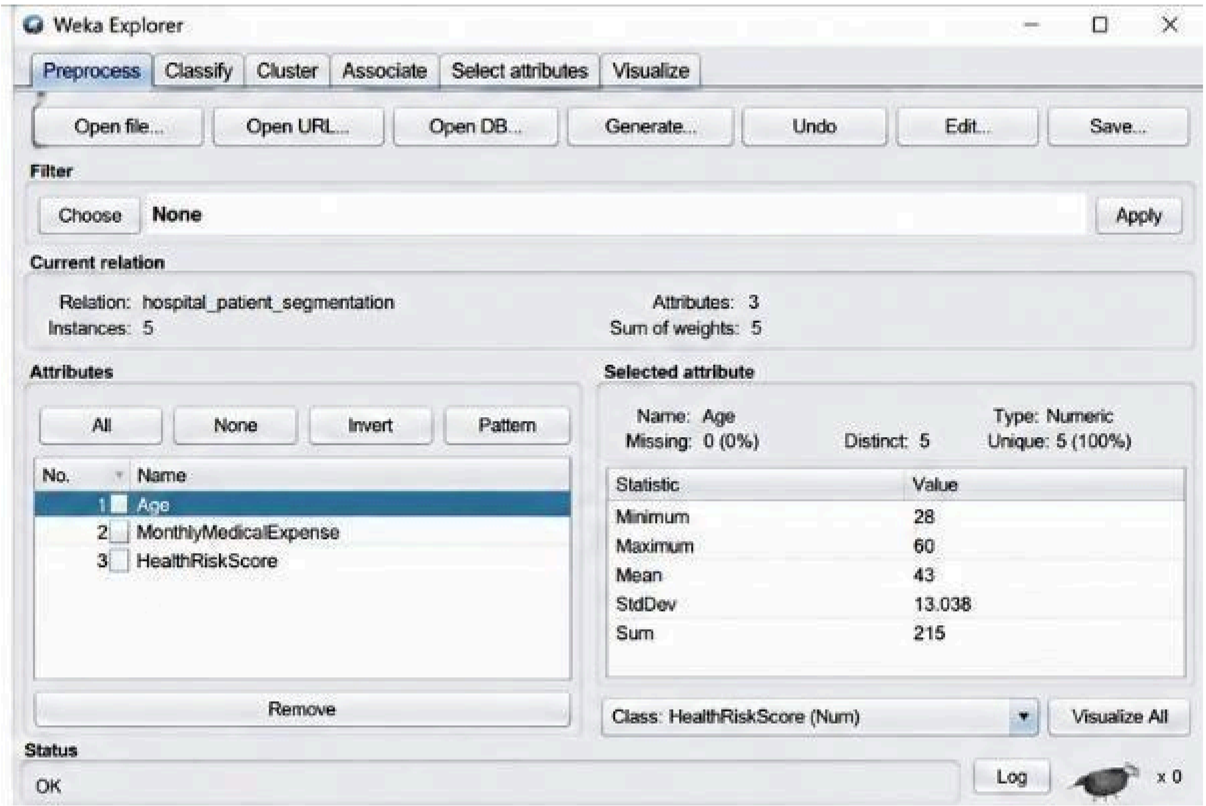
Type: Nominal
Missing: 0 (0%)
Distinct: 5
Unique: 5 (100%)
No. Label Count Weight
1 P1 1 1.0
2 P2 1 1.0
3 P3 1 1.0
4 P4 1 1.0
5 P5 1 1.0
Class: HealthRiskScore (Num)
Visualize All
Status
OK
Log
x 0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Open DB...
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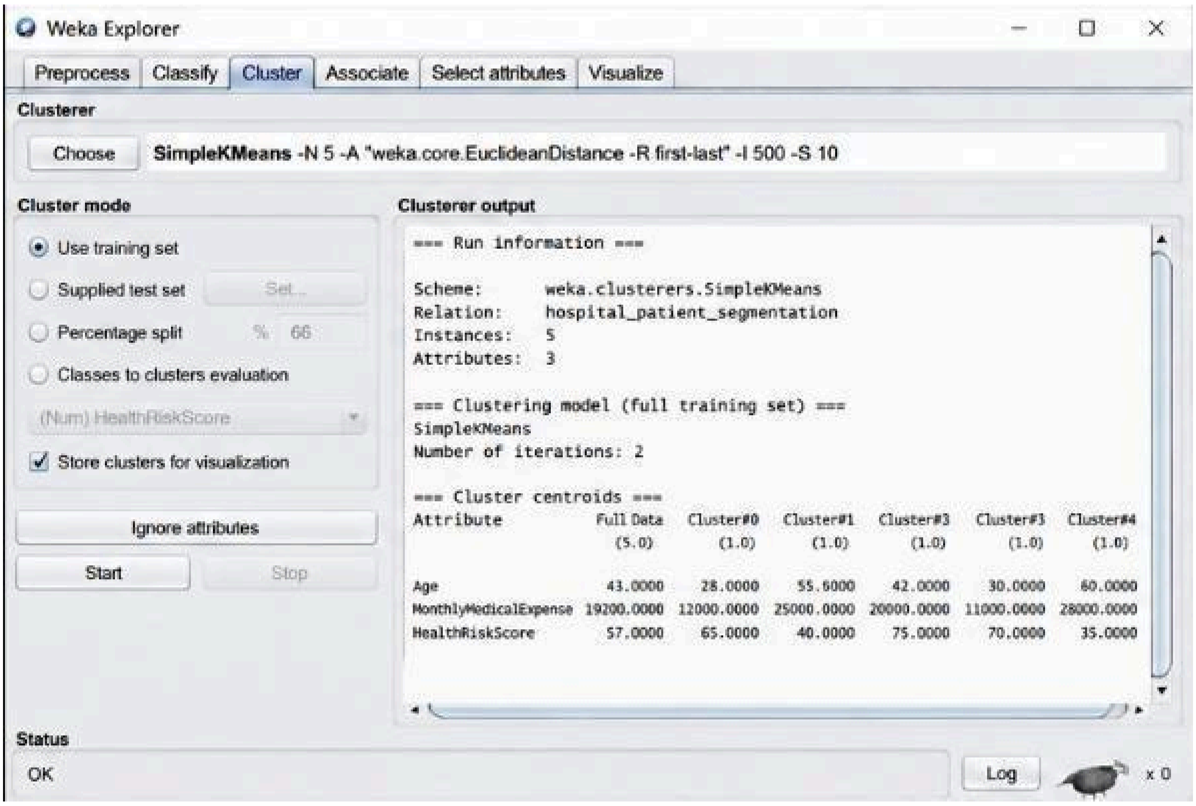
Save...
Filter
Choose
None
Apply
Current relation
Relation: hospital_patient_segmentation
Instances: 5
Attributes: 3
Sum of weights: 5
Attributes
All
None
Invert
Pattern
No. Name
1 Age
2 MonthlyMedicalExpense
3 HealthRiskScore
Remove
Selected attribute
Name: Age
Type: Numeric
Missing: 0 (0%)
Distinct: 5
Unique: 5 (100%)
Statistic Value
Minimum 28
Maximum 60
Mean 43
StdDev 13.038
Sum 215
Class: HealthRiskScore (Num)
Visualize All
Status
OK
Log
x 0


Weka Explorer
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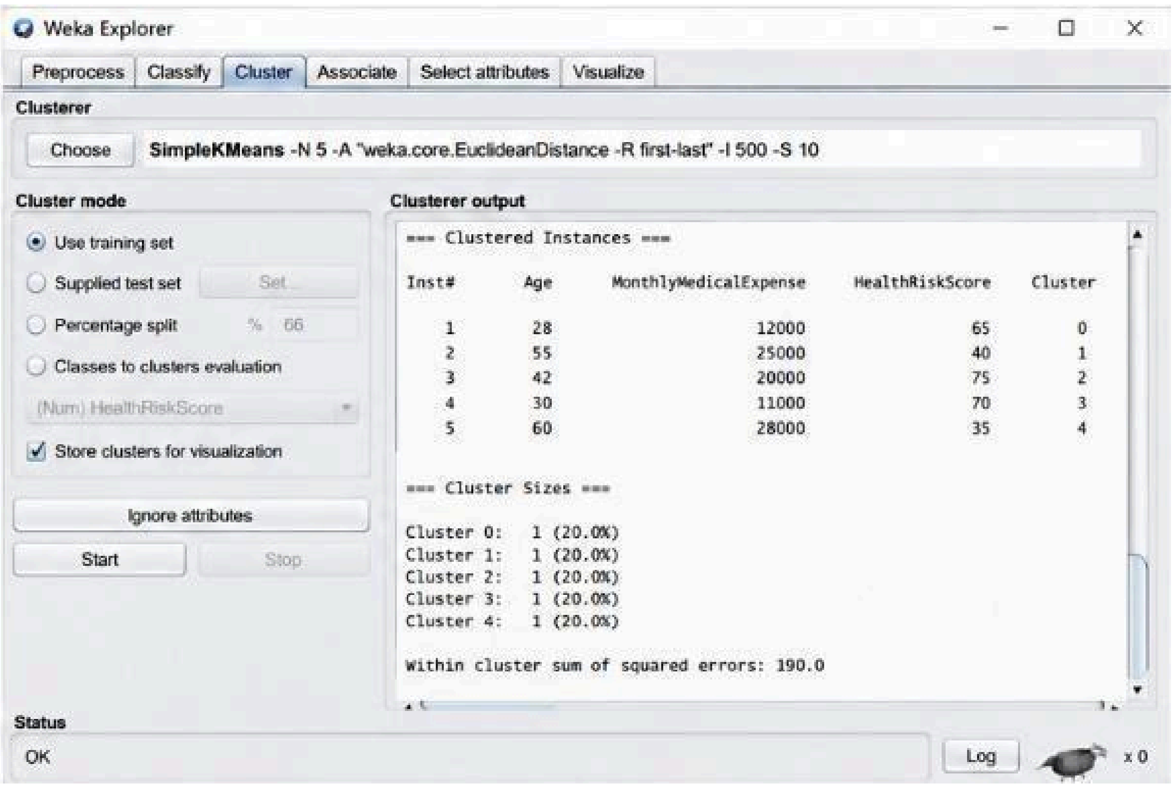
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
SimpleKMeans -N 5 -A "weka.core.EuclideanDistance -R first-last" -I 500 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) HealthRiskScore
Store clusters for visualization
Ignore attributes
Start
Stop
Clusterer output
=== Run information ===
Scheme: weka.clusterers.SimpleKMeans
Relation: hospital_patient_segmentation
Instances: 5
Attributes: 3
=== Clustering model (full training set) ===
SimpleKMeans
Number of iterations: 2
=== Cluster centroids ===
Attribute Full Data Cluster#0 Cluster#1 Cluster#3 Cluster#3 Cluster#4
(5.0) (1.0) (1.0) (1.0) (1.0) (1.0)
Age 43.0000 28.0000 55.0000 42.0000 30.0000 60.0000
MonthlyMedicalExpense 19200.0000 12000.0000 25000.0000 20000.0000 11000.0000 28000.0000
HealthRiskScore 57.0000 65.0000 40.0000 75.0000 70.0000 35.0000
Status
OK
Log
x 0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Choose
SimpleKMeans -N 5 -A "weka.core.EuclideanDistance -R first-last" -I 500 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) HealthRiskScore
Store clusters for visualization
Ignore attributes
Start
Stop
Clusterer output
=== Clustered Instances ===
Inst# Age MonthlyMedicalExpense HealthRiskScore Cluster
1 28 12000 65 0
2 55 25000 40 1
3 42 20000 75 2
4 30 11000 70 3
5 60 28000 35 4
=== Cluster Sizes ===
Cluster 0: 1 (20.0%)
Cluster 1: 1 (20.0%)
Cluster 2: 1 (20.0%)
Cluster 3: 1 (20.0%)
Cluster 4: 1 (20.0%)
Within cluster sum of squared errors: 190.0
Status
OK
Log
x 0

Q2.E-CommerceOrderAnalysis
Answer
Hierarchicalclusteringgroupssimilarobservationstogetherbasedontheircharacteristics.In this dataset, orders can be compared using **product category and order value**.
Theordervalues are:

- O1–Electronics–₹55,000
- O2– Clothing– ₹25,000
- O3–Electronics–₹57,000
- O4–Furniture– ₹30,000
- O5– Clothing– ₹27,000

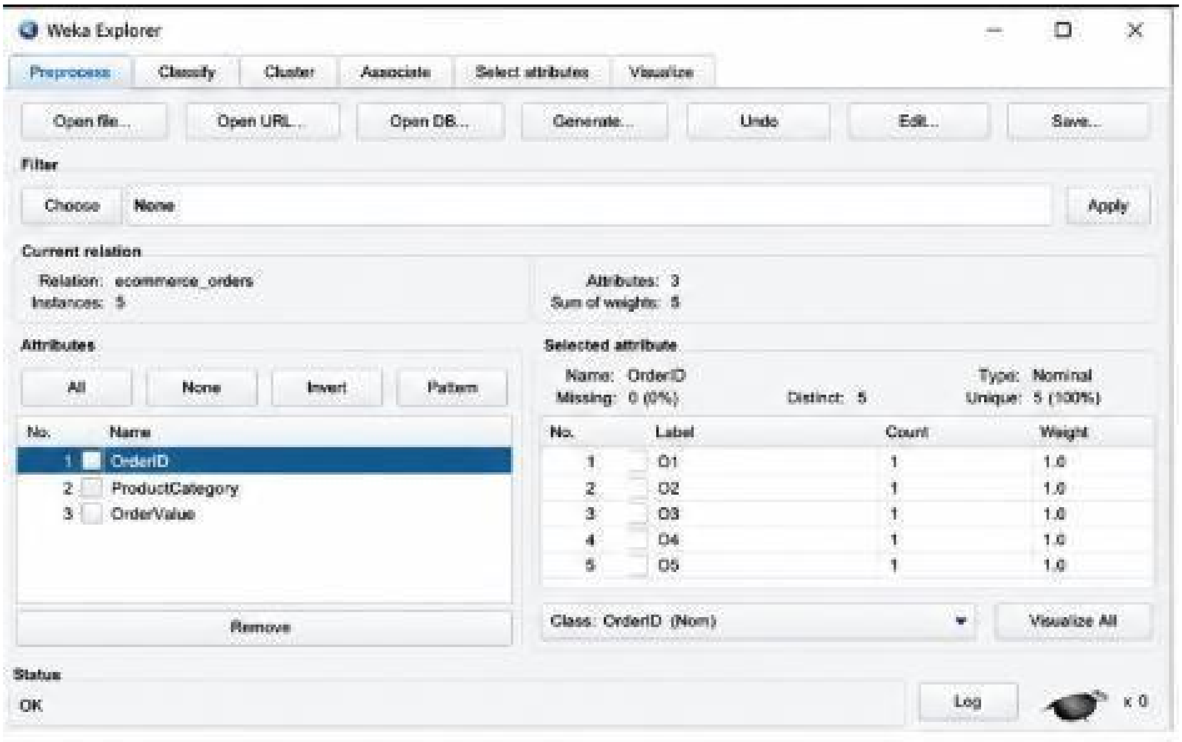
Initially, each order is considered as an individual cluster. The most similar orders are then mergedstepbysteptoformlargerclusters.Adendrogramrepresentsthesemergingoperations.
Cluster Formation
Asuitableinterpretation is:

- **Cluster1:**O1andO3–Electronicsorderswithverysimilarhigh ordervalues.
- **Cluster2:**O2andO5–Clothingorderswithsimilarordervalues.
- **Cluster3:**O4– Furnitureorder,whichdiffersfromtheothergroups.

Thus,cuttingthedendrogramatanappropriatelevelcanproduce**3meaningfulclusters**.
Significance
Theclustershelptheretailerunderstandpurchasingbehavior:

- TheElectronicsclusterrepresentscustomersplacing**high-valueelectronicsorders**.
- TheClothingclusterrepresents**medium-valueclothingpurchases**.
- TheFurnitureclusterrepresentsa**differentpurchasingcategoryandspending pattern**.

Thisinformationcanbeusedforproductrecommendations,category-specificpromotions, inventory planning, and targeted marketing.
Conclusion
Asuitablesolutionis**3clusters**,becausethegroupsrepresentclearlydifferentpurchasing patterns and product categories.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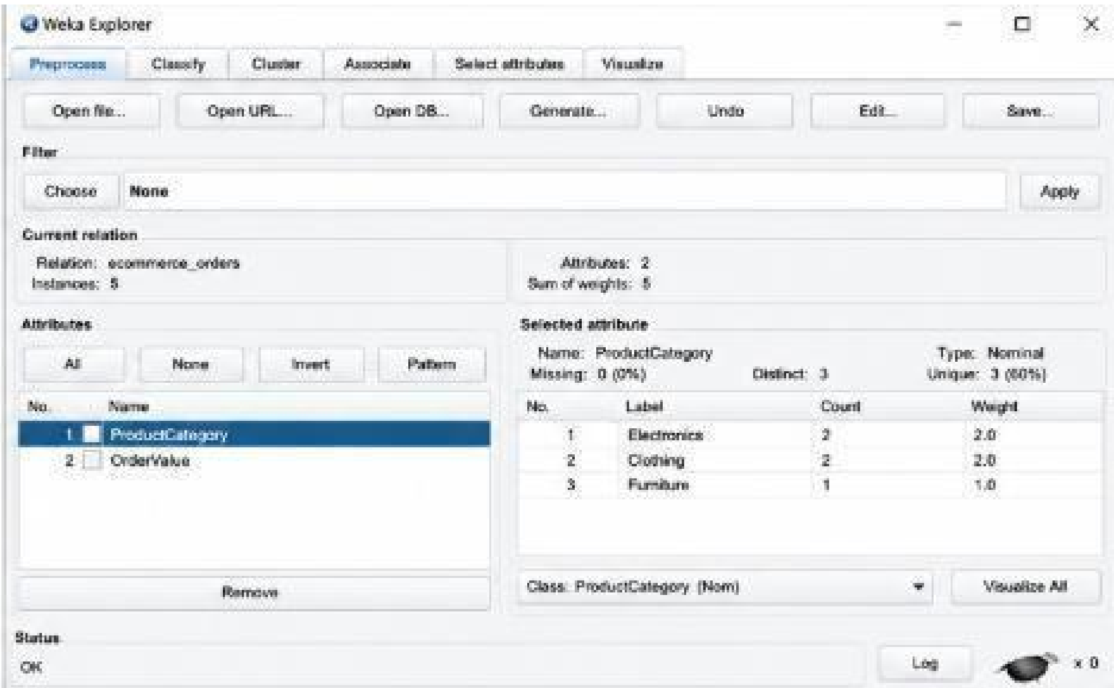
Filter
Choose
None
Apply
Current relation
Relation: ecommerce_orders
Instances: 5
Attributes: 3
Sum of weights: 5
Attributes
All
None
Invert
Pattern
No. Name
1 OrderID
2 ProductCategory
3 OrderValue
Remove
Selected attribute
Name: OrderID
Missing: 0 (0%)
Distinct: 5
Type: Nominal
Unique: 5 (100%)
No. Label Count Weight
1 O1 1 1.0
2 O2 1 1.0
3 O3 1 1.0
4 O4 1 1.0
5 O5 1 1.0
Class: OrderID (Nom)
Visualize All
Status
OK
Log
x 0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Filter
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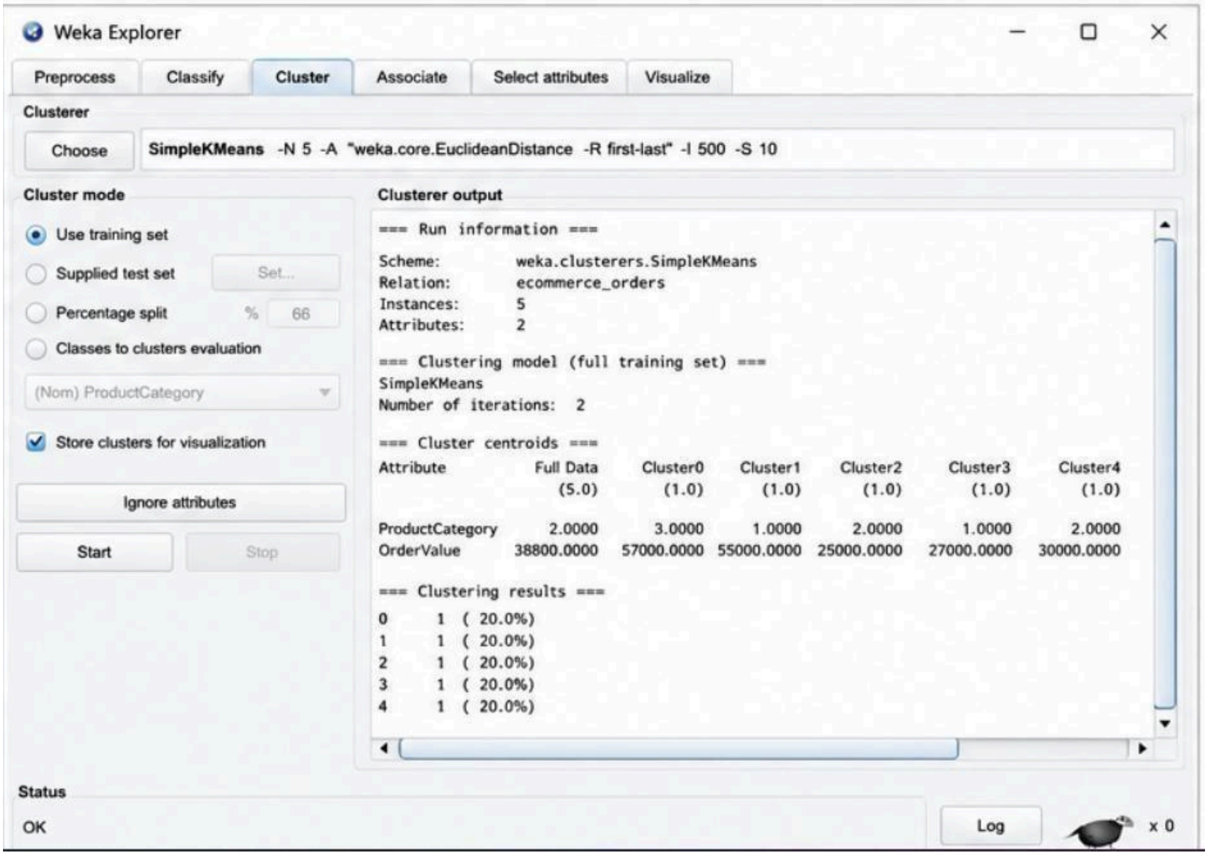
Apply
Current relation
Relation: ecommerce_orders
Instances: 5
Attributes: 2
Sum of weights: 5
Attributes
All
None
Invert
Pattern
No. Name
1 ProductCategory
2 OrderValue
Remove
Selected attribute
Name: ProductCategory
Missing: 0 (0%)
Distinct: 3
Type: Nominal
Unique: 3 (60%)
No. Label Count Weight
1 Electronics 2 2.0
2 Clothing 2 2.0
3 Furniture 1 1.0
Class: ProductCategory (Nom)
Visualize All
Status
OK
Log
x 0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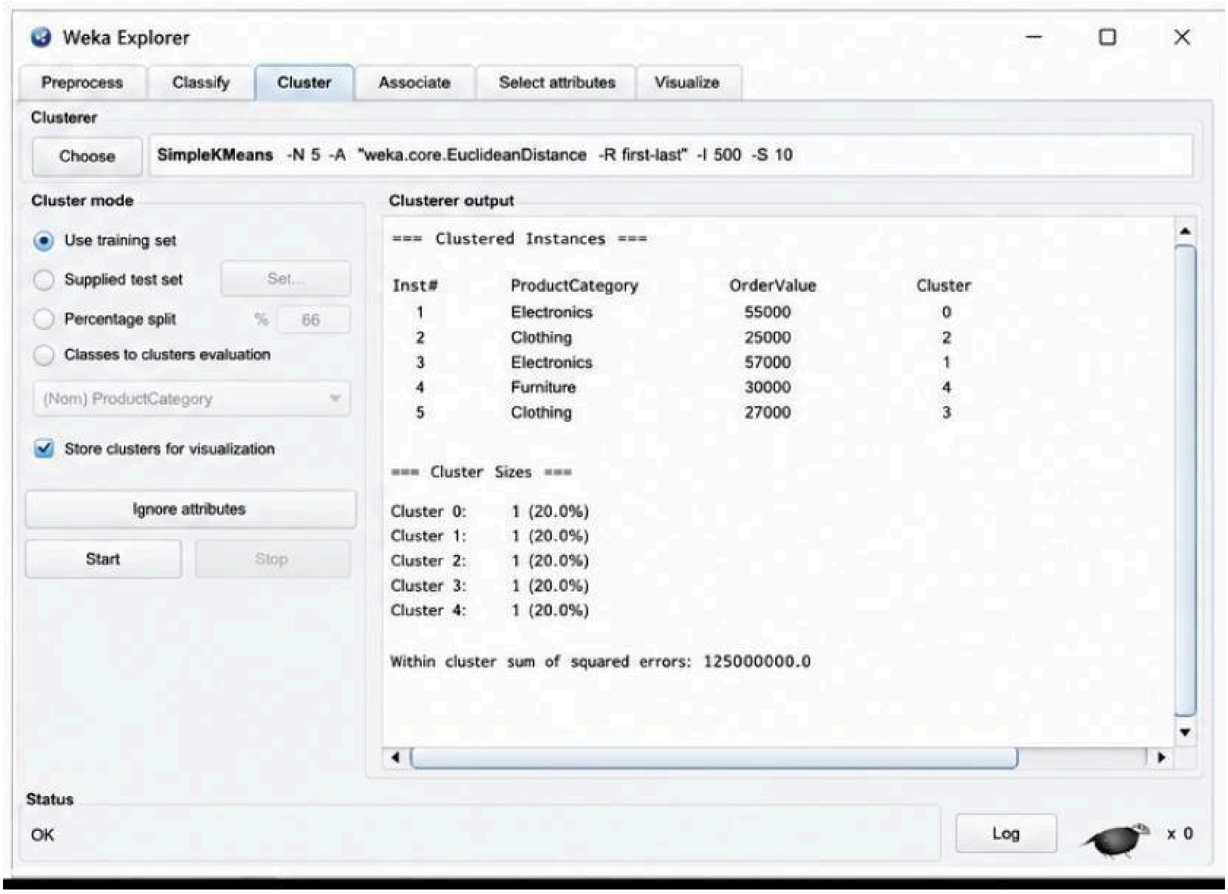
Choose
SimpleKMeans -N 5 -A "weka.core.EuclideanDistance -R first-last" -I 500 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Nom) ProductCategory
Store clusters for visualization
Ignore attributes
Start
Stop
Clusterer output
=== Run information ===
Scheme: weka.clusterers.SimpleKMeans
Relation: ecommerce_orders
Instances: 5
Attributes: 2
=== Clustering model (full training set) ===
SimpleKMeans
Number of iterations: 2
=== Cluster centroids ===
Attribute Full Data Cluster0 Cluster1 Cluster2 Cluster3 Cluster4
(5.0) (1.0) (1.0) (1.0) (1.0) (1.0)
ProductCategory 2.0000 3.0000 1.0000 2.0000 1.0000 2.0000
OrderValue 38800.0000 57000.0000 55000.0000 25000.0000 27000.0000 30000.0000
=== Clustering results ===
0 1 (20.0%)
1 1 (20.0%)
2 1 (20.0%)
3 1 (20.0%)
4 1 (20.0%)
Status
OK
Log
x 0



Q3.MobileAppUserBehavior Answer

Thecompanyusesclusteringtodivideusersaccordingtotheir**dailyusagetime,login frequency, and in-app purchases**.

Theusershavedifferentactivitypatterns:

User	**UsageTime**	**LoginFrequency**	**Purchase**
U1	120	18	₹2,000
U2	60	5	₹5,000
U3	140	22	₹1,500
U4	50	4	₹6,000
U5	130	20	₹1,800

UsersU1,U3andU5havehighusagetimeandhighloginfrequency.Therefore,theyare likely to form a **high-engagement user group**.

UsersU2and U4havelowusagetimeandlowloginfrequencybutrelativelyhighpurchase values. They can form a **low-engagement but high-spending group**.

Probability-BasedMembership

Insoftclustering,auserisnotforcedtobelongcompletelytoonlyonecluster.Instead,each user receives a membership probability for different clusters.

Forexample:

User	HighEngagement	LowEngagement/High Spending
U1	0.90	0.10
U2	0.20	0.80
U3	0.95	0.05
U4	0.10	0.90
U5	0.92	0.08

Thesevaluesareillustrativemembershipvalues.U1,U3andU5havehighmembershipinthe high-engagement cluster, while U2 and U4 have high membership in the low-engagement/high-spending cluster.

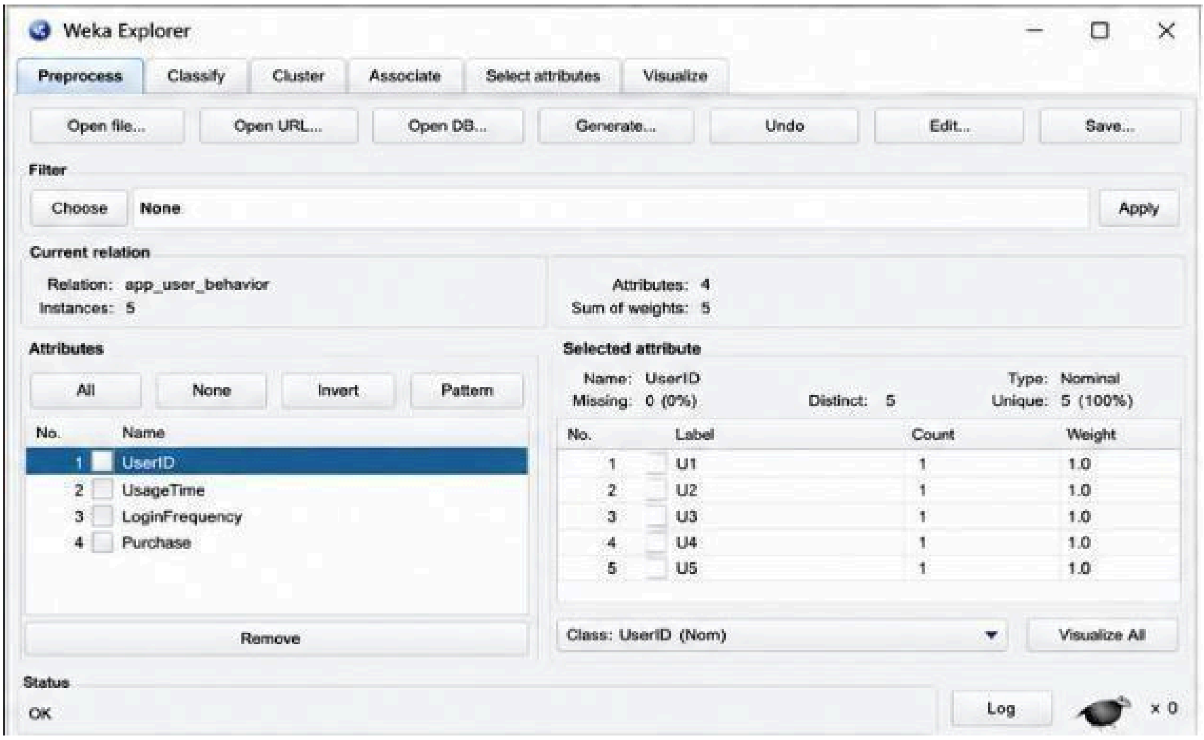
AdvantagesofSoftClustering

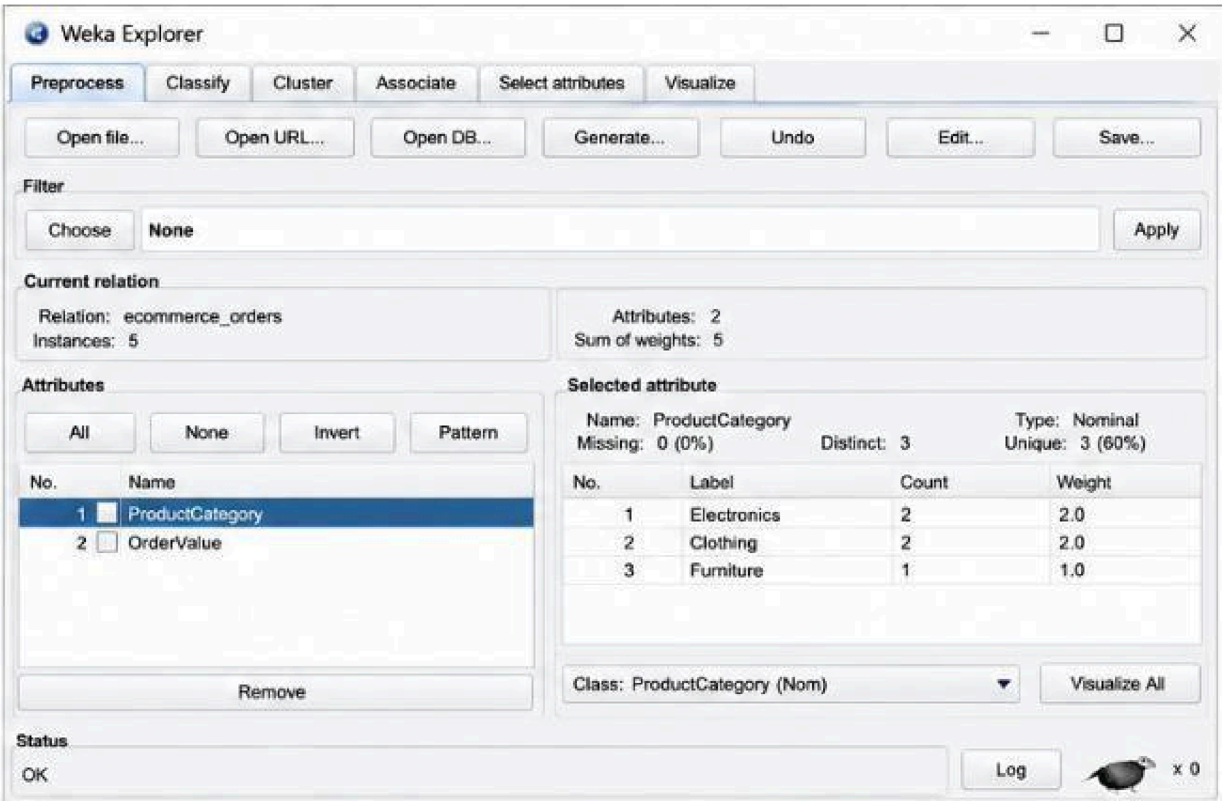
Softclusteringisuseful because:

1. Ausercanpartially belongtomultiplebehavioralgroups.
2. Itrepresentsuncertaintymorerealistically.
3. Usersnearclusterboundariescanbeidentified.
4. Marketingstrategiescanbepersonalizedaccordingtomembershipprobabilities.
5. Ithelpsidentifyuserswhomaychangefromonebehaviorgroupto another.

Conclusion

Softclusteringprovidesmoreflexiblecustomersegmentationthanhardclusteringbecauseit captures overlapping user behavior and uncertainty.




Weka Explorer
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Select attributes
Visualize
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Current relation
Relation: ecommerce_orders
Instances: 5
Attributes: 2
Sum of weights: 5
Attributes
All
None
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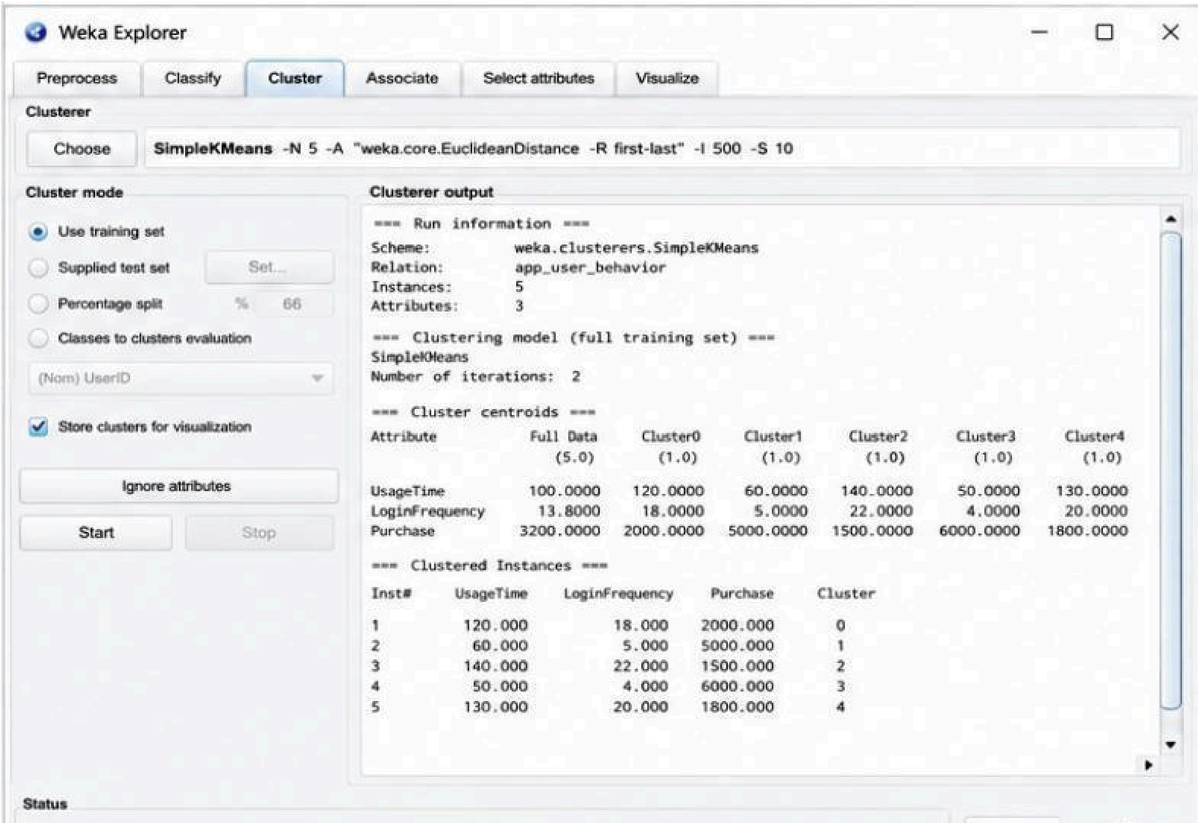
Pattern
No. Name
1 ProductCategory
2 OrderValue
Remove
Selected attribute
Name: ProductCategory
Type: Nominal
Missing: 0 (0%)
Distinct: 3
Unique: 3 (60%)
No. Label Count Weight
1 Electronics 2 2.0
2 Clothing 2 2.0
3 Furniture 1 1.0
Class: ProductCategory (Nom)
Visualize All
Status
OK
Log
x 0


Weka Explorer
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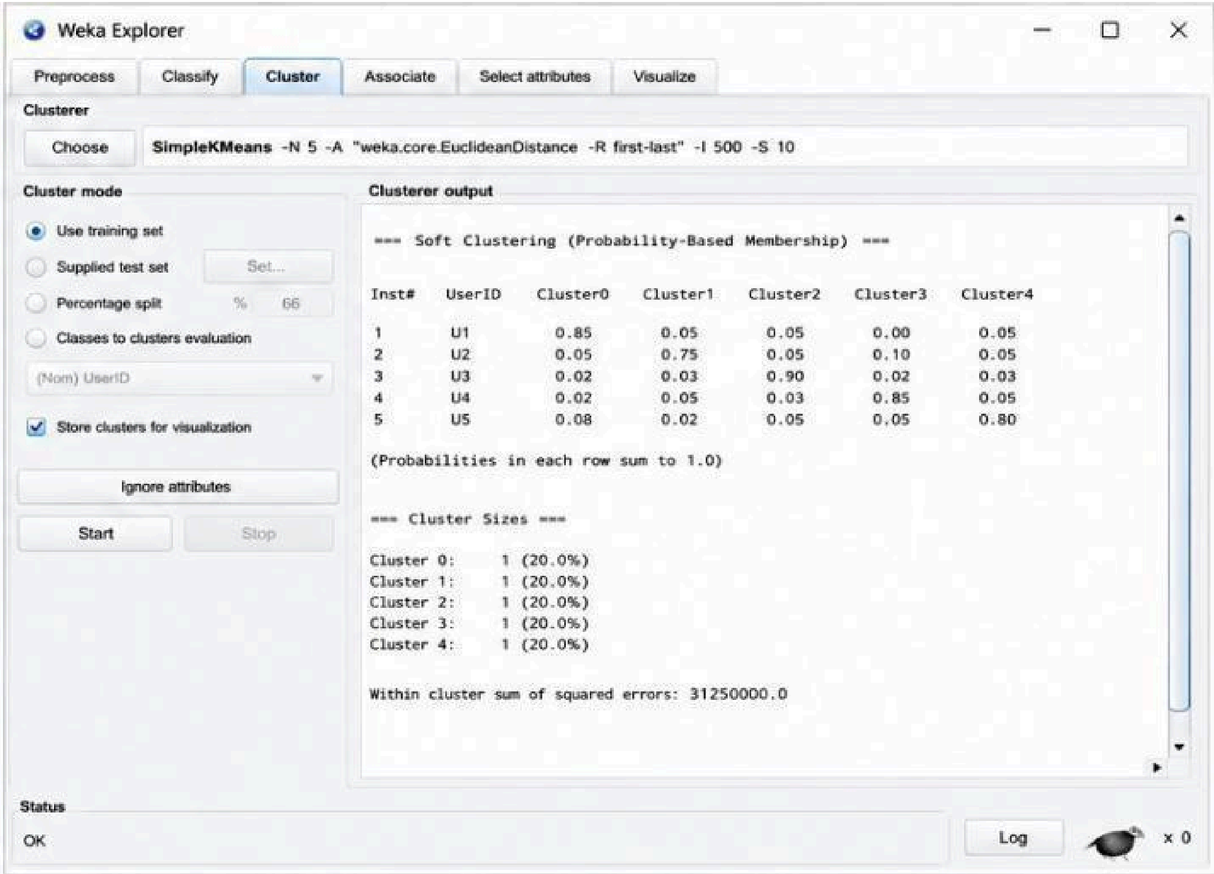
Clusterer
Choose
SimpleKMeans -N 5 -A "weka.core.EuclideanDistance -R first-last" -I 500 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Nom) UserID
Store clusters for visualization
Ignore attributes
Start
Stop
Clusterer output
=== Run information ===
Scheme: weka.clusterers.SimpleKMeans
Relation: app_user_behavior
Instances: 5
Attributes: 3
=== Clustering model (full training set) ===
SimpleKMeans
Number of iterations: 2
=== Cluster centroids ===
Attribute Full Data Cluster0 Cluster1 Cluster2 Cluster3 Cluster4
(5.0) (1.0) (1.0) (1.0) (1.0) (1.0)
UsageTime 100.0000 120.0000 60.0000 140.0000 50.0000 130.0000
LoginFrequency 13.8000 18.0000 5.0000 22.0000 4.0000 20.0000
Purchase 3200.0000 2000.0000 5000.0000 1500.0000 6000.0000 1800.0000
=== Clustered Instances ===
Inst# UsageTime LoginFrequency Purchase Cluster
1 120.000 18.000 2000.000 0
2 60.000 5.000 5000.000 1
3 140.000 22.000 1500.000 2
4 50.000 4.000 6000.000 3
5 130.000 20.000 1800.000 4
Status



Q4.AgriculturalCropMonitoring

Answer

Theagriculturaldatasetcontainsthreeattributes:

- Farmrainfall
- Soilfertility score
- Cropyield

Theattributeshavedifferentscales.Rainfallrangesapproximatelyfrom**280–520mm**,soil fertility ranges from **3–9**, and crop yield ranges from **65–90 tons**.

BeforeNormalization

Without normalization, the rainfall values have a much larger numerical range than soil fertility.Distance-basedclusteringalgorithmsmaythereforegiveexcessiveimportanceto rainfall. Asaresult,farmsmaybegroupedmainlyaccordingtotheirrainfallvaluesratherthan considering all three agricultural characteristics equally.

Forexample:

- F1andF4haverelatively lowrainfallandhighsoilfertility.
- F2andF5havehighrainfall andlowsoil fertility.
- F3hasintermediatevalues.

AfterNormalization

Normalization converts the attributes into comparable scales, commonly between **0 and 1**. After normalization, rainfall, soil fertility, and crop yield contribute more equally to the distancecalculation.Therefore,theclustersrepresentoverallagriculturalsimilarityinsteadof being dominated by rainfall.

Possiblegroupsinclude:

- **Highfertilityandhighyieldfarms:**F1andF4
- **Highrainfallandlowerfertility/yieldfarms:**F2andF5
- **Intermediateagriculturalconditions:**F3

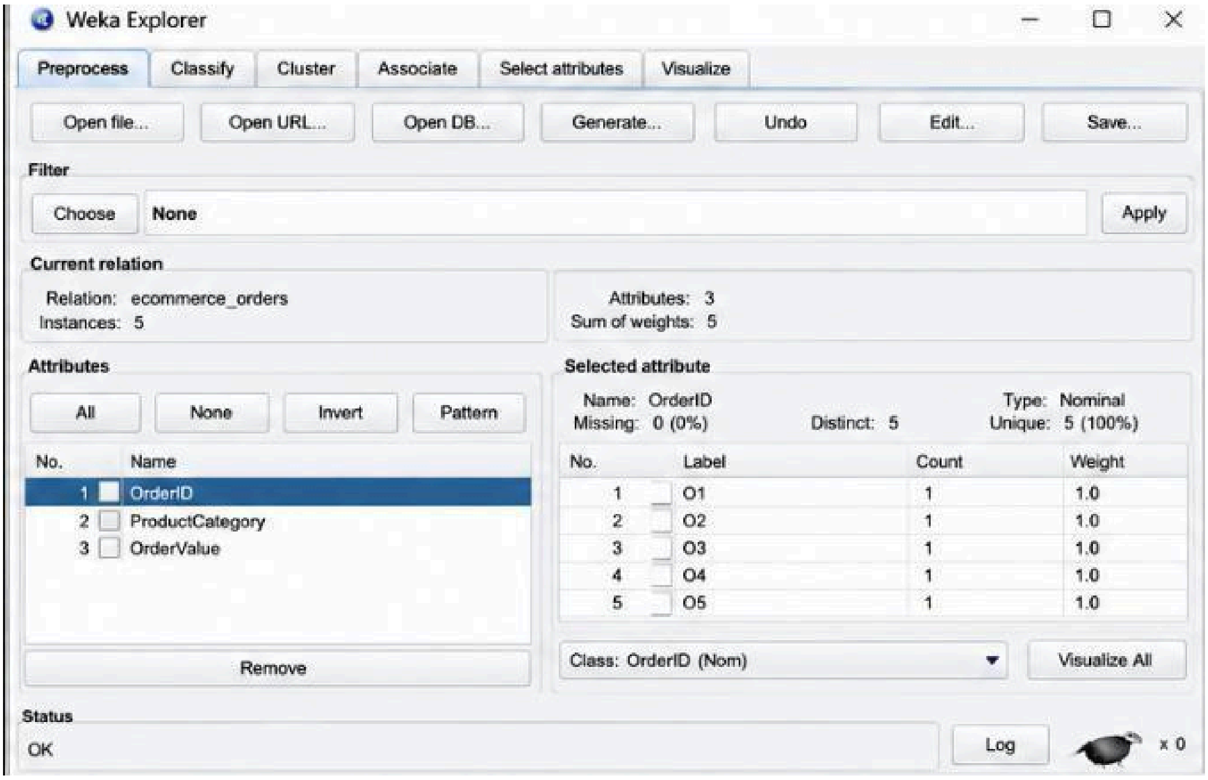
ImportanceofNormalization

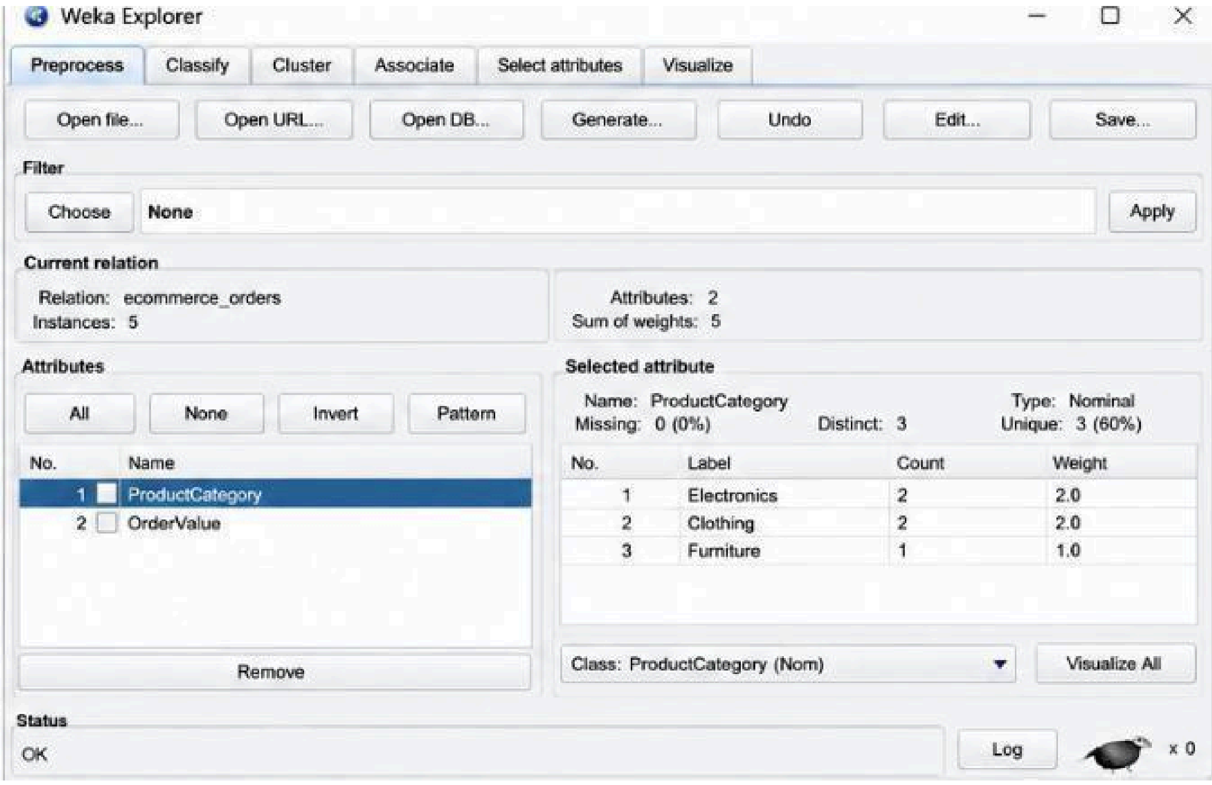
Normalization:

1. Preventslarge-scaleattributesfromdominatingclustering.
2. Giveseachfeatureafaircontribution.
3. Producesmoremeaningfulclusters.
4. Improvesthereliabilityofdistance-basedalgorithms.
5. Helpsresearchersmakebetteragriculturaldecisions.

Conclusion

Normalizationisimportantbecauseitensuresthatallagriculturalattributescontributefairlyto cluster formation. Therefore, the clusters obtained after normalization are generally more reliable and meaningful.

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Attributes: 2
Sum of weights: 5
Attributes
All
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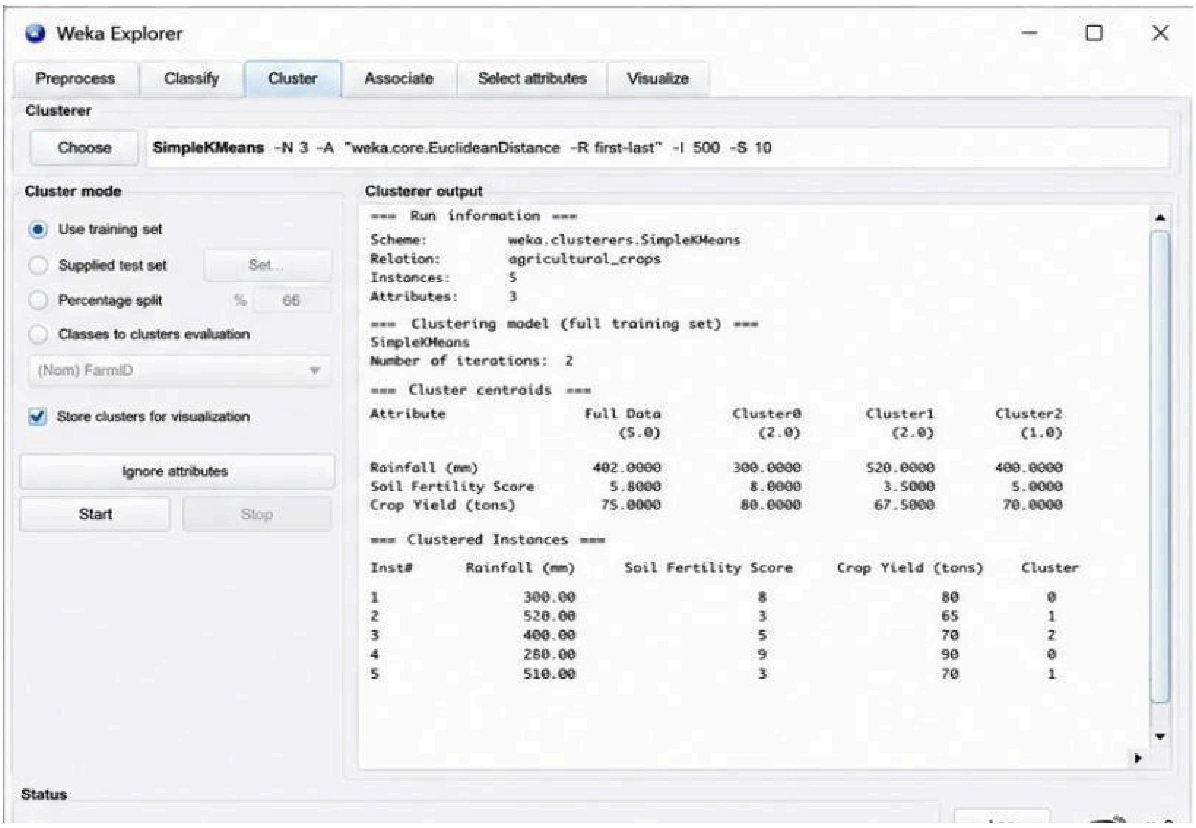
Invert
Pattern
No. Name
1 ProductCategory
2 OrderValue
Remove
Selected attribute
Name: ProductCategory
Type: Nominal
Missing: 0 (0%)
Distinct: 3
Unique: 3 (60%)
No. Label Count Weight
1 Electronics 2 2.0
2 Clothing 2 2.0
3 Furniture 1 1.0
Class: ProductCategory (Nom)
Visualize All
Status
OK
Log
x 0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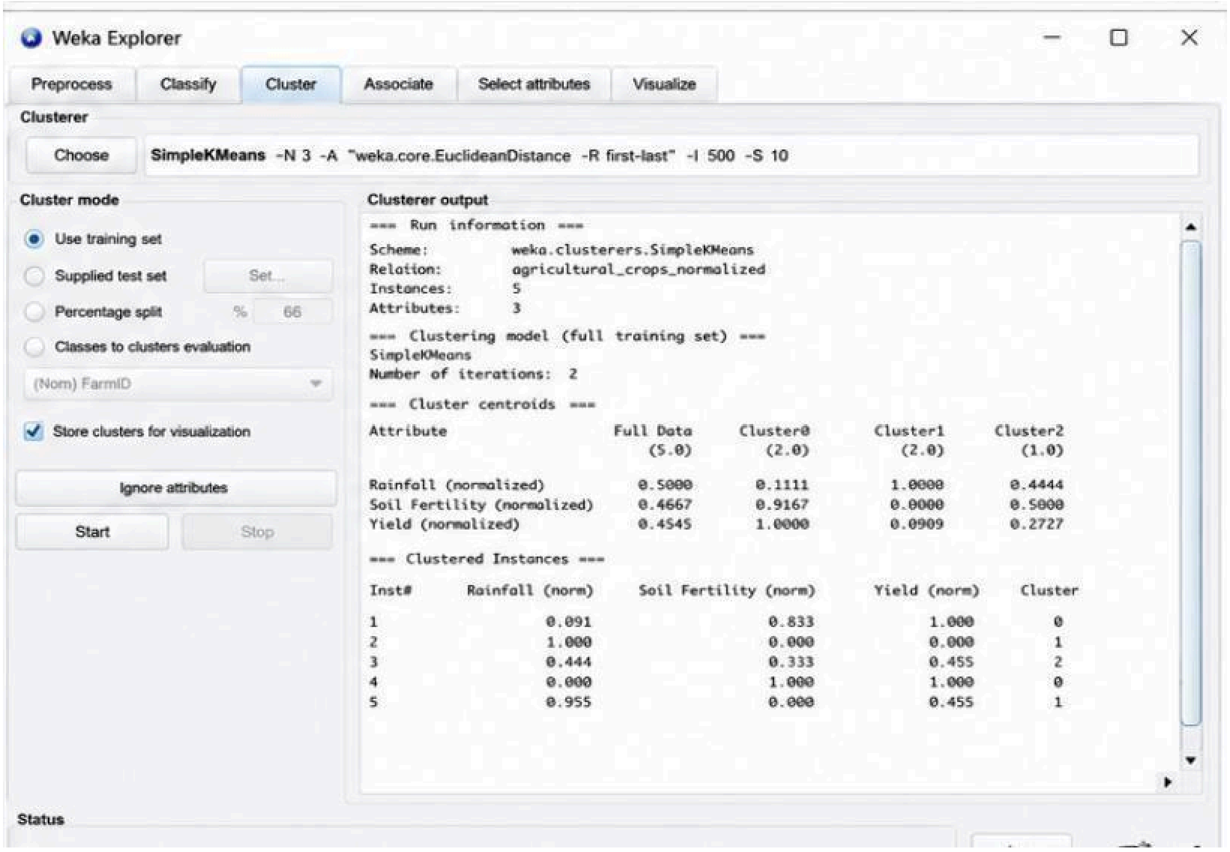
SimpleKMeans -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Nom) FarmID
Store clusters for visualization
Ignore attributes
Start
Stop
Clusterer output
=== Run information ===
Scheme: weka.clusterers.SimpleKMeans
Relation: agricultural_crops
Instances: 5
Attributes: 3
=== Clustering model (full training set) ===
SimpleKMeans
Number of iterations: 2
=== Cluster centroids ===
Attribute Full Data Cluster0 Cluster1 Cluster2
(5.0) (2.0) (2.0) (1.0)
Rainfall (mm) 402.0000 300.0000 520.0000 400.0000
Soil Fertility Score 5.8000 8.0000 3.5000 5.0000
Crop Yield (tons) 75.0000 80.0000 67.5000 70.0000
=== Clustered Instances ===
Inst# Rainfall (mm) Soil Fertility Score Crop Yield (tons) Cluster
1 300.00 8 80 0
2 520.00 3 65 1
3 400.00 5 70 2
4 280.00 9 90 0
5 510.00 3 70 1
Status



Q5.EmployeePerformanceEvaluation Answer

Thegivensilhouettescores are:

NumberofClusters	**SilhouetteScore**
K=2	0.48
K=3	**0.68**
K=4	0.52

Asilhouettescoremeasureshowwelldatapointsfitwithintheirownclustercomparedwith other clusters. A **higher silhouette score indicates better-defined and more separated clusters**. Amongthethreemodels:

- K=2hasascoreof**0.48**.
- K=3hasthehighest scoreof**0.68**.
- K=4hasascoreof**0.52**.

Therefore,**K=3isthemostsuitablenumberofclusters**. **Employee Segmentation**

Thethreeclusterscanrepresentdifferentperformancegroups:

- **Cluster1–Highperformers:**Highproductivity,attendance,andskillscores.
- **Cluster2–Averageperformers:**Moderateproductivity,attendance,andskillscores.
- **Cluster 3 – Lower performers:** Lower productivity, attendance, and skill scores. For example, E1 and E5 have high scores and are likely to belong to the high-performance group.E4hasrelativelylowscoresandmaybelongtothelower-performancegroup.E2and E3 may represent the middle-performance group.

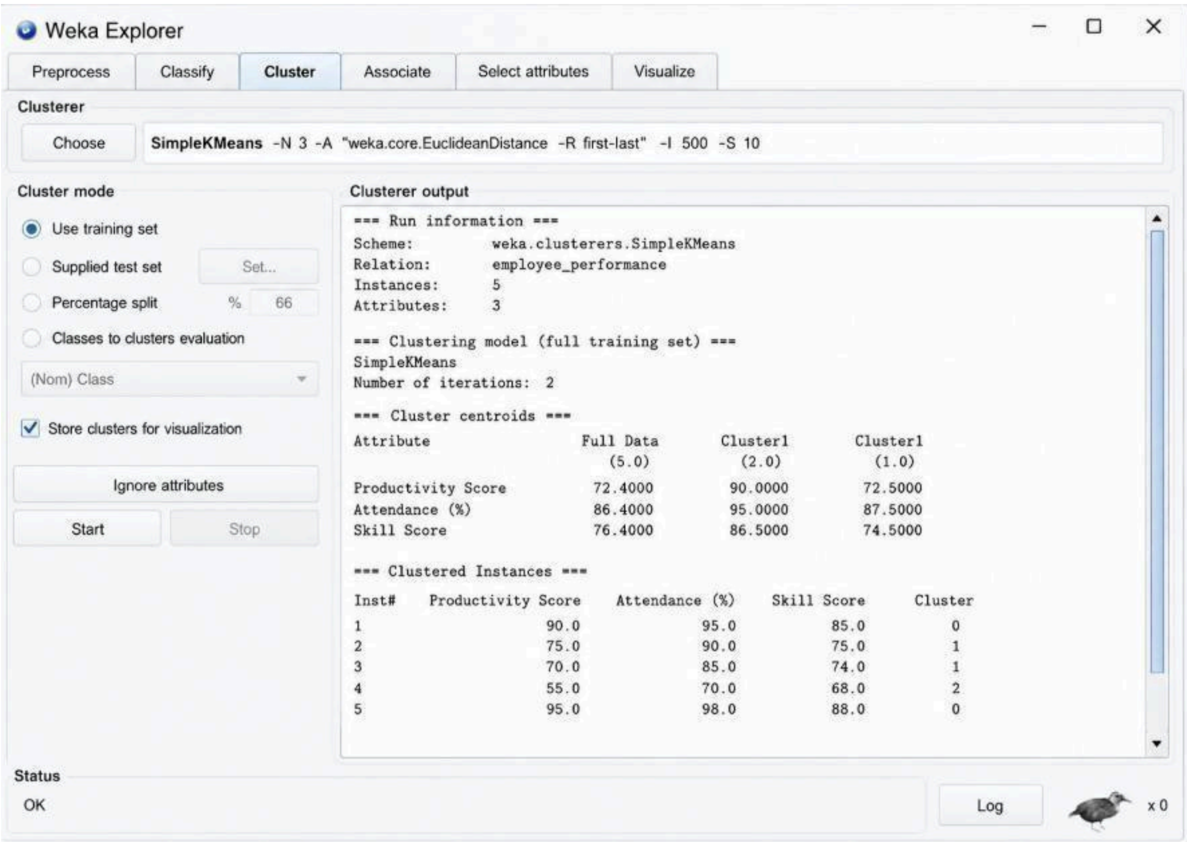
ImportanceforEmployeeEvaluation

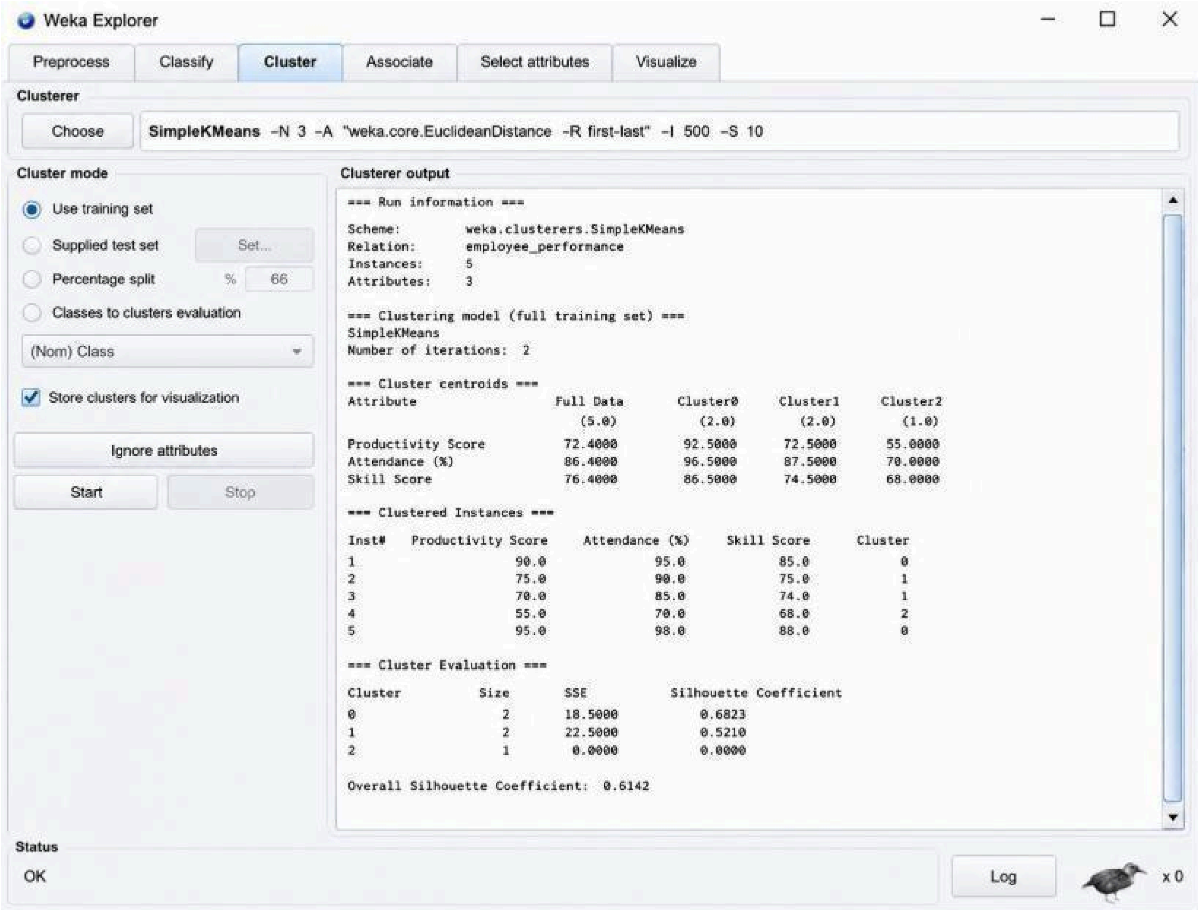
Usingthreeclustershelpsthecompany:

1. Identifyhigh-performingemployees.
2. Identifyemployeeswho mayneedtraining.
3. Understanddifferentperformancepatterns.
4. Planrewardsandrecognition.
5. Designtargetedemployeedevelopmentprograms.

Conclusion

Thebest modelis**K = 3 becauseithasthehighestsilhouettescoreof 0.68**. Itprovides better-definedemployee groupsthanK=2orK=4 andcansupporteffectiveperformance evaluation and workforce management.




Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
SimpleKMeans -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Nom) Class
Store clusters for visualization
Ignore attributes
Start
Stop
Clusterer output
=== Run information ===
Scheme: weka.clusterers.SimpleKMeans
Relation: employee_performance
Instances: 5
Attributes: 3
=== Clustering model (full training set) ===
SimpleKMeans
Number of iterations: 2
=== Cluster centroids ===
Attribute Full Data Cluster0 Cluster1 Cluster2
(5.0) (2.0) (2.0) (1.0)
Productivity Score 72.4000 92.5000 72.5000 55.0000
Attendance (%) 86.4000 96.5000 87.5000 70.0000
Skill Score 76.4000 86.5000 74.5000 68.0000
=== Clustered Instances ===
Inst# Productivity Score Attendance (%) Skill Score Cluster
1 90.0 95.0 85.0 0
2 75.0 90.0 75.0 1
3 70.0 85.0 74.0 1
4 55.0 70.0 68.0 2
5 95.0 98.0 88.0 0
=== Cluster Evaluation ===
Cluster Size SSE Silhouette Coefficient
0 2 18.5000 0.6823
1 2 22.5000 0.5210
2 1 0.0000 0.0000
Overall Silhouette Coefficient: 0.6142
Status
OK
Log
x 0